Mikhail Razakov

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Software Engineer · .NET, backend, and systems

I design and implement complex software systems: modular .NET platforms, backend services with payments and recovery, compilers, and program-analysis tools. My strongest areas are explicit contracts, state invariants, verifiable semantics, and carrying systems through to reproducible outcomes.

Platform architecture
UniversalToolchain/Wist2: independent packages, typed contracts, routes, manifests/locks, and lifecycle ownership

Production backend
Payments, subscriptions, roles, devices, PostgreSQL/SQLite, migrations, recovery, backup/restore, and rollback

Compiler / systems
Callable-first SSA, interpreter/CIL, LLVM, C++23, memory analysis, and x86-64 code generation

EXPERIENCE

2024 — present

UniversalToolchain / Wist2 — creator and primary developer
- Designed language composition from independently versioned packages: typed artifacts, capabilities and conflicts, deterministic routes, and pass ordering.
- Implemented exact manifest binding, lock identity, fail-closed policies, and runtime-session lifecycle ownership.

July 1 — August 31, 2026

MCST — compiler engineering intern
- Developing an LLVM optimization pass and implementing tested graph algorithms in C++23.

2023 — 2025

Mama Lama — Software Developer
- Independently developed games and applications for DJs and events, from requirements and design through implementation, testing, and preparation for use.
- Designed the application architecture and modular structure, defining component boundaries and interfaces; owned the complete technical implementation through a finished product.

PROJECTS

UniversalToolchain / Wist2
The current exact test manifest passes with zero failures; interpreter/CIL parity, clean-consumer scenarios, and reproducible compiler/runtime contracts are verified.

VpnMediator
A production service with controlled state transitions, backup/restore, health gates, and safe rollback; the public case study excludes secrets and user data.

PS-form Memory Dependence Analyzer
Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

SKILLS

Architecture
module boundaries, typed contracts, state machines, manifests/locks, lifecycle ownership, fail-closed design

.NET / backend
C#/.NET, ASP.NET Core, REST/OpenAPI, PostgreSQL, SQLite, webhooks, transactions, migrations, recovery

Compilers
Bytecode/AIR, CFG, SSA, dominance, optimizations, interpreter/CIL, LLVM, program analysis

Systems and verification
C++23, C17, Rust, Python, Linux, CMake, sanitizers, differential/metamorphic testing, exact oracles

EDUCATION

Software Engineering student at HSE University

RECOGNITION

The LangDev 2026 Program Committee accepted a talk on the architecture of UniversalToolchain/Wist2 for presentation in Torremolinos, Spain.

Moscow / remote